

exactly 80 mm: _____

FIRST SOLVE

New here? Do the walkthrough once at cubepath.52.vercel.app/!
 This article is the memory jog.

NOTATION

R U F L D B = turn that face clockwise, looking at it from outside the cube.
 ' = counter-clockwise, 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together.
 y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns · trigger = a short algorithm your hands run at as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.
 Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RUR'U'



white sticker FRONT
L'ULU



white sticker UP run the first one once to knock it out, then look again
RUR'U'

3 MIDDLE EDGES

Top edge with NO yellow, turn U until its front color matches the center under it. The arrow shows which slot it goes to.



goes RIGHT
U RUR'U' y L'ULU



goes LEFT
U' L'ULU y' L'ULU' RUR'U'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ RUR'U' sexy move
■ RUR'U' Sune
■ FRF' sledgehammer
gap = change grip

FIRST SOLVE

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R U F L D B = turn that face clockwise, looking at it from outside the cube.
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algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

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goes RIGHT
U' R'UR' y L'ULU



goes LEFT
U' L'ULU y' RUR'U'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

RUR'U' sexy move
 RUR'U Sune
 R'RFR' sledgehammer
 gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line far left-to-right
FRUR'U'F'



Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{3}$ yellow, rest clockwise
RUR'U RU2R'



Anti-Sune $\frac{1}{3}$ yellow, rest counter-clockwise
RU2R' U'RUR'



Pi $\frac{2}{3}$ yellow, headlights left
FRUR'U'f' FRUR'U'F'

OLL CORNERS - continued



Headlights $\frac{2}{3}$ yellow, headlights at the back
R2UR'U2 RD'R'U2R'



2x Headlights $\frac{2}{3}$ yellow, headlights both sides
RUR'U RU'R' URU2R'



Chameleon $\frac{2}{3}$ yellow next to each other
rUR'U'F' FR'



Bowtie $\frac{2}{3}$ yellow diagonal
F'FRUR'U' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, three is the worst. Turn not all yellow and nothing matches? Run Sunes and read again.

No yellow edge at the end? Run the first one, then read again.

RUR'U wavy move, **RUR'U** Sunes, **R'FR'** single algorithm, **sun** = change erin

2-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
fRUR'U'f'

Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow; rest clockwise
RUR'U' RU2R'

Anti-Sune $\frac{1}{2}$ yellow; rest counter-clockwise
RU2R' U'RUR'

Pi $\frac{1}{2}$ yellow; headlights left
fRUR'U'f' fRUR'U'f'

OLL CORNERS - continued



Headlights $\frac{3}{4}$ 2 yellow; headlights at the back
R2R'U2R' RD'R'U2R'

2x Headlights $\frac{3}{4}$ 0 yellow; headlights both sides
RUR'U' RU'R' URU2R'

Chameleon $\frac{3}{4}$ 2 yellow; next to each other
rUR'U'r' f'f'

Bowtie $\frac{3}{4}$ 2 yellow diagonal
F'rUR'U'r' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn, also finish the case. Machine-checked, there is the worst.
Top not all yellow and nothing matches? Run Sunes and read again.

No yellow edge at Run the first one then read again.

RUR'U' = easy move, **RUR'U'U'** = 2 moves, **fRUR'U'f'** = 3 moves

sun = 2 moves, **sun** = 3 moves, **sun** = 4 moves

ONE-LOOK PLI

ADJACENT CORNER SWAP

exactly one face shows headlights



- T1 headlights: pairs B-left F-left
RUR'U' R'F RZUR'UR' RUR'F'
- Ja L solid: pairs B-right F-left B-front
x RZFRF' RU2 rUr U2
- Jb L solid: pairs B-left F-right B-back
RUR'F' RUR'U' R'F RZUR'U'
- Aa L headlights: pairs F-right B-front
x LZD2 L'UL D2 L'UL'
- Ab L headlights: pairs B-right B-back
x' LZD2 LUL' D2 LU'L
- F1 L solid
R'U'F' RUR'U' R'F RZUR'U' RUR'UR

One face shows headlights; two corners of that face match. Hold as shown, turn, then turn top to finish.

RUR'U' sec.moves, **RUR'U'** Sym.Sym. **R'FR'** slideschamber **erao** = change **eri**

ADJACENT CORNER SWAP

continued



- Ra L1 headlights: pairs F-right
RUR'U' RU' RU'R UR'D' R'U'2R'
- Rb L1 headlights: pairs B-left
R2F RUR'U' FR U2RU'2R
- Ga L1 headlights: pairs F-right
RZUR'UR'U' RU'R2 U'D R'UR'D'
- Gb L1 headlights: pairs B-back
RUR'U' UD' RZUR'UR'U' RU'R2D
- Gc L1 headlights: pairs B-right
RZUR'UR'U' RU'R2 UD' RU'R'D
- Gd L1 headlights: pairs B-front
RUR'U' UD' RZUR'UR'U' UR'UR'2D

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ONE-LOOK PLI

ADJACENT CORNER SWAP

exactly one face shows headlights

• T1 headlights: pairs R-left F-left
RUR'U' RF R2UR'U' RUR'F'

• Ja1 solid: pairs B-right F-left R-front
x R2FRF' RU2 rUr U2

• Jb1 solid: pairs B-left F-right R-back
RUR'F' RUR'U' R'F R2UR'

• Aa1 headlights: pairs F-right B-front
x L2D2 L'U'L D2 L'U'L'

• Ab1 headlights: pairs B-right R-back
x' L2D2 LUL' D2 LU'L

• F1 solid
RU'F' RUR'U' RF R2UR'U'
RUR'UR

One face shows two corners of that face match. Hold as shown, then turn the top to finish.

■ **RURV** activate ■ **RURU** turn ■ **RF'RF'** sliderchange **can** = change grip

ADJACENT CORNER SWAP

continued

• Ra1 headlights: pairs F-left
RUR'U' RUR DR'UR'D' RU'2R'

• Rb1 headlights: pairs B-left
R2F RURU' FR U2RU'2R

• Ga1 headlights: pairs F-right
R2UR'UR'U' RU'R2 U'D R'UR'D'

• Gb1 headlights: pairs R-back
RUR' UD' R2UR'URU' RU'R2D

• Gc1 headlights: pairs B-right
R2UR'URU' RU'R2 UD' RU'R'D

• Gd1 headlights: pairs R-front
RUR' UD' R2UR'URU' UR'XUR'D

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

[illegible][illegible]

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the SHORT edge. The numeral on both faces is your check: cut ONE card first. This bar must measure exactly 80 mm:

[illegible]